

Florian Handke

Head of Industrial Security · Campus Schwarzwald · M.Sc. Mechanical Engineering

Phone: +49 151 553 603 39 | **Email:** handke.florian@web.de | **LinkedIn:** [linkedin.com/in/florian-handke](https://www.linkedin.com/in/florian-handke)

Head of Industrial Security at Campus Schwarzwald, a cluster of companies in the mechanical engineering and manufacturing industries. Holding a Master's degree (M.Sc.) in Mechanical Engineering with a specialization in Production Engineering, I am responsible for ensuring the security of industrial machinery and components. My expertise focuses on Public Key Infrastructure (PKI), digital identities, and their applications in industrial environments.

Core Expertise

Applied Cryptography

- Public Key Infrastructure (PKI)
- Certificate Lifecycle Management
- Device Identity & Secure Device Identity
- Secure Boot & Secure Firmware Updates
- Code Signing
- Hardware Security Modules (HSMs)
- PKCS#11 API Implementation (Python)
- PKCS#x Standards

Industrial Security

- IEC 62443 Compliance
- Cyber Resilience Act (CRA) Assessment
- NIS2 Directive
- Security Architecture & Risk Assessment
- Industrial IoT & OT Security
- Expertise in IEEE 802.1 AR device identities (IDeVID & LDeVID)
- Expertise in zero-touch onboarding (BRSKI, FDO)

IoT & Connectivity

- OPC UA (Implemented GDS Push)
- MQTT Protocol
- Grafana & Visualization
- Telegraf Data Collection
- InfluxDB Time-Series Database
- Node-RED (Published 2 NPM packages)
- Mosquitto MQTT Broker

Software Engineering

- Python (Django, uv)
- Shell Scripting
- Docker & Containerization
- GitHub Actions & CI/CD
- CodeQL & Static Code Analysis
- SBOM (CycloneDX, SPDX)
- OpenSSF Best Practices & Badges
- OWASP ZAP Security Testing

Featured Project

Trustpoint – Open Source Trust Anchor

Co-created and technically led the development of **Trustpoint**, an open-source trust anchor and PKI platform. Trustpoint manages digital machine identities and certificate lifecycles in industrial environments, supporting secure device onboarding, certificate issuance, renewal, revocation, and decommissioning. It operates locally in segmented, offline, or air-gapped OT networks.

GitHub: github.com/Trustpoint-Project/trustpoint

Professional Experience

Sept. 2023 – Present · 2 years 11 months

Head of Industrial Security

Campus Schwarzwald

Jan. 2021 – Aug. 2023 · 2 years 8 months

Smart Production Engineer

Campus Schwarzwald

Apr. 2018 – Dec. 2020 · 2 years 9 months

Data Scientist: Condition Monitoring of Laser Devices

TRUMPF

Oct. 2015 – Sept. 2016 · 1 year

Customer and Project Manager

Andreas Lupold Hydrotechnik GmbH

Education

2016 – 2018

Master of Science (M.Sc.), Advanced Precision Engineering

Hochschule Furtwangen

2012 – 2015

Bachelor of Engineering (B.Eng.), Production Engineering

Duale Hochschule Baden-Württemberg

2011 – 2012

University of Applied Sciences Entrance Qualification

Berufskolleg Horb a.N.

2007 – 2011

Apprenticeship as a Precision Mechanic

Koch Pac Systeme GmbH

Leadership & Community

Technical Leadership

- Led multidisciplinary development teams
- Architected security solutions
- Technical mentoring & motivation
- Disciplinary Management of Employees
- Proposal writing (German & EU Horizon) — Last one (EU Horizon) called PQCUA (sadly not accepted): Implementing Post-Quantum Cryptography into OSS OPC UA SDKs (open62541 and S2OPC)

Teaching & Public Speaking

- Part-time lecturer at DHBW Stuttgart (Industrial Security)
- Part-time lecturer at University of Stuttgart (Industrial Security)
- Industrial security trainings & workshops (Applied Cryptography Training)
- Conference speaker & presenter
- Author of technical articles (ATP, embedded world conference, IT-SICHERHEIT, IT&Produktion)

Community & Events

- VDMA Industrial Security Working Group
- Open Industry 4.0 Working Group
- Organizer: PKI in der Automatisierungstechnik (since 2023)
- Organizer: OT-Security Forum
- Organizer: OT-Security Breakfast

Selected Talks & Keynotes

- **Automated Identity Management for Embedded Components** — embedded world conference, March 2026
- **Industrial trust anchor** — all about automation, March 2026
- **Digital Identities for the industry** — SPS Messe, November 2025
- **Industrial Security at Campus Schwarzwald** — OT Security Breakfast, Oktober 2025
- **Cryptography and Legal Requirements for Businesses** — Campus Schwarzwald, Mai 2025
- **Certificate Management Made Easy – Practical Solutions for Machinery and Plant Operators** — Hannover Messe, April 2025
- **Securing Industry 4.0 with Digital Identities for Embedded Components** — embedded world conference, March 2025
- **The rising importance of managed certificates in industrial systems** — embedded world conference, March 2025
- **From Emmental to Gouda: Why Cybersecurity Is Now a Management Issue** — Business Breakfast, February 2025
- **Digital trust** — Campus Cyber Paris, December 2024
- **How PKI can help you meet the Cyber Resilience Act** — PKI in der Automatisierungstechnik, September 2024
- **Digital identities for a secure industry** — 20. Deutscher IT-Sicherheitskongress, April 2024
- **Why Trust is the foundation for the connected factory** — Cyber Security Symposium, March 2024
- **Building Trust in the Factory Through Digital Identities** — RWTH Aachen, March 2024
- **Digital Identities for Machines—How Does It Really Work?** — Security unter Kontrolle, September 2023

About Me

- I combine engineering pragmatism with technical expertise in applied cryptography and industrial cybersecurity. With a background in mechanical engineering, I understand how the industry works. I then moved into software and security, gaining in-depth knowledge through hands-on experience.
- I thrive on taking ownership, leading technical teams, speaking at conferences, teaching students and engineers, and building solutions that solve real-world problems. I love details, thinking end-to-end, and presenting complex content to different audiences. I specialize in turning complex security concepts into practical engineering solutions that balance technical feasibility with cost-effectiveness—because security is always a tradeoff.
- I live in a small village in the Black Forest, Baden-Württemberg, with my wife and two daughters.
- I play football as a hobby.
- Hands-on engineer who loves taking action and building things.
- Every good concept starts with a sequence diagram.
- Good user experience matters—even (especially!) in security software.
- The world has lost a carpenter and an electrician in me—I'm an expert in building trade show exhibits.
- I probably have the right adapter for your plug.